Algebra 1 Name____________________
Unit 6 Date____________________
Lesson 2 Homework Period____________________

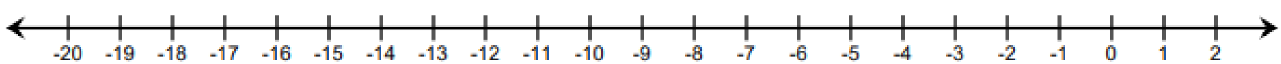
Use the following conjunctive inequality to answer questions 1-3.

$$5v + 10 \leq -4v - 17 < 9 - 2v$$

Rewrite the compound inequality as two single inequalities and solve each one.

1.	2.

3. Graph the solution set of the compound inequality.

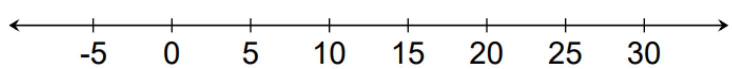


Kali just started a new sales floor job to save for college. She earns $\$15.75$ per hour plus a flat fee of $\$50$ per week. She wants to earn between $\$200$ and $\$400$ per week. The following inequality represents her earning potential:

$$200 \leq 15.75x + 50 \leq 400$$

4. Solve the inequality.

5. Graph the solution on the number line. Approximate values as necessary.



6. Complete the statement: Kali can work between _____ and _____ hours each week.

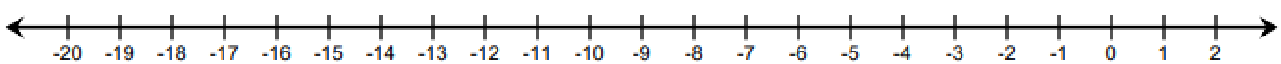
Use the disjunctive inequality shown below to answer questions 7-10:

$$-\frac{15}{4}\left(2x-\frac{2}{3}\right) < 64+\frac{11}{4}x \quad or \quad \frac{35x-11}{4} \leq -134$$

Solve each component of the compound inequality.

7. $-\frac{15}{4}\left(2x-\frac{2}{3}\right) < 64+\frac{11}{4}x$	8. $\frac{35x-11}{4} \leq -134$

9. Graph the solution sets of both inequalities on the number line.



10. Explain which of the following values is not a solution to the inequality.

-20 $\quad -10$ $\quad 0$